

Fig. 2: Pin configurations of LM7805, UM66 and BC547

which is further given to IC3. These binary codes at data pins 2, 3 and 4 of IC3 ensure that at least one of its output pins—13, 14 or 15—is high. This makes the output pin 6 of OR gate (N1) high, which performs three tasks:

1. Provides power supply to seven-segment display DIS1 to display caller's number through transistor T3.
2. Blocks the acceptance of further input due to low voltage at EI pin 5 of encoder IC2, as transistor T1 conducts.
3. Provides input power supply to IC4 to produce sound to alert the attendant.

After the attendant responds and

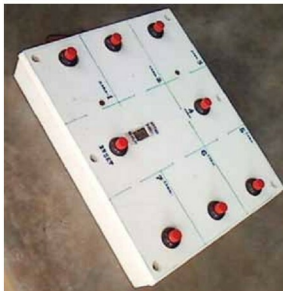


Fig. 3: Author's prototype

presses reset switch S8 momentarily, all the outputs of IC2 go low. This makes the output of OR gate IC4 low. This, in turn, results in cutting off the power supply to the seven-segment display and melody generator IC6. The encoder IC2 is permitted to accept next input through switches S1 through S7

by providing high voltage to its EI pin 5.

Assemble the circuit on a general-purpose PCB. Refer Fig. 2 for pin configurations of ICs 7805, UM6 and BC547. After assembling the circuit, enclose it in a suitable cabinet along with reset switch S8, DIS1 and speaker, and place it in the attendant's room. Extend all switches, S1 through S8, to respective rooms using two-wire cables for each switch. Fix the transformer inside the cabinet. Fix power-on LED1 at front panel of the cabinet. Connect a power

cable to the primary of the transformer for providing the mains supply to the circuit.

Alternatively, if you wish to demonstrate the circuit as a project, the switches can be fixed on the the top of enclosure as shown in author's prototype (Fig. 3). ●

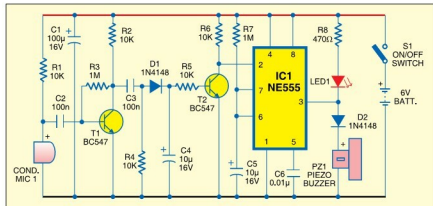
SOUND SENSOR ALARM

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This 6V battery-operated circuit triggers an acoustic piezobuzzer when a sound is detected. It can also be used as a cheap acoustic-

type glass break detector and/or ambient sound level monitor.

The circuit has an ordinary condenser microphone MIC1 as a sound sensor. Sensitivity of this microphone can be changed to some extent by



changing the value of the bias resistor R1. When the circuit is powered by a 6V battery through switch S1, it goes into standby mode and the red LED1 lights up to indicate that the circuit is ready for use.

When microphone (MIC) detects a sound, electrical signal from the microphone is amplified and processed by a small-signal amplifier wired around transistor T1 (BC547). Amplified signal from the collector of T1 is passed to electrolytic capacitor C4 through diode D1 (1N4148). Transistor T2 (BC547) conducts and triggers the monostable built around the timer IC NE555 (IC). As a result,